

3GPP STANDARDS & ARCHITECTURAL SUMMARY

Joint Synthesis of Highlights Issue 08 (June 2024) & Technical Report 28.908 (AI/ML Management)

1. Evolution Timeline & Core Release Parameters

The 3GPP ecosystem is managing a multi-tiered structural transition across overlapping standard configurations. The system lifecycle dictates simultaneous progression across several major cellular technological thresholds:

- **Release 18 (The 5G-Advanced Baseline):** Achieved operational maturity in mid-2024 with Stage 3 functional freezes completed by Technical Specification Group Core Network and Terminals (TSG CT). It acts as the initial blueprint for cross-layer Optimization frameworks.
- **Release 19 (Bridge to Next Generation Structural Units):** Curates network-wide intelligence infrastructure. Major operational parameters target Functional Freezes spanning late 2025 across key working groups including RAN 2/3/4 and SA5 Management & Orchestration workflows.
- **Release 20 (Formal 6G Strategic Inception):** Allocates multi-month Project-Wide 6G Structural Studies starting across 2025–2026. This release outlines primary requirements for IMT-2030 Use Cases and defines the structural topology for 6G network cores.

2. 6G Strategic Framework & Global Policy Integration

Formalization parameters for IMT-2030 (6G) were anchored during the Stage-1 Workshop on Use Cases in Rotterdam. The global technical strategy addresses six distinct multi-dimensional capabilities:

IMT-2030 Core Capability	Structural Target & Network Purpose
Immersive Communication	Directs ultra-high-definition interactive rendering, spatial mapping, and multi-sensorial digital twin synchronization.
Integrated Sensing & Comms	Copurposes the active cellular interface to support spatial localization, weather tracking, and physical motion detection.
Ubiquitous Connectivity	Establishes universal mesh-coverage integrating deep terrestrial footprints with space-based Non-Terrestrial Networks (NTN).
AI-Native Networking	Embeds direct computation matrices within standard layers to automate real-time localized waveform adjustments.
Sustainability & Circularity	Enforces strict energy-saving operational benchmarks across multi-vendor equipment loops to slash systemic carbon footprints.

3. Vertical Utility Integration & Smart Grid Interoperability

Under Release 18 specifications, the 3GPP framework formally introduces standardized cross-industry coordination interfaces via the **Network Services and Operations for Energy Utilities (NSOEU)** specification layer (TS 28.318). Historically, Distribution Service Operators (DSOs) relied on crude over-the-top monitoring which lacked granularity and caused extended localized outages during major power failures.

NSOEU mandates a standardized, bi-directional information exchange protocol between Mobile Network Operators (MNOs) and DSOs. By exposing bounded performance metadata (Uplink/Downlink Packet Loss, Average Cell Latency, and Absolute Network Availability), DSOs can execute proactive network re-routing within seconds instead of resetting physical routers (which introduces multi-minute baseline downtime). Furthermore, NSOEU communicates real-time infrastructure metrics: MNO essential sites transmit their exact autonomous backup energy capacities to DSOs, allowing grid systems to prioritize substation automated restoration sequences before base station backup fuel arrays are exhausted.

Cross-Layer Multimedia Frameworks & AI Lifecycle Management

4. Immersive XR & Real-Time Communication Technology Layers

To normalize immersive multimedia transport across structurally diverse user terminal categories, 3GPP has deployed an explicit **XR Baseline Client UE Architecture** within TS 26.119. This framework bridges application logic directly to the 5G network subsystem:

- **Device Category Mapping:** Disaggregates terminal profiles into four explicit computational groups: Thin AR Glasses (highly power-constrained), Rich AR Glasses (high localized execution budgets), XR Phones (smartphone ecosystem integration), and XR Head-Mounted Displays (HMDs).
- **Structural Codec Pipeline:** Enforces standardized execution configurations utilizing Immersive Voice and Audio Services (IVAS) codecs alongside ISO/IEC gLTF 2.0 scene spatial mapping architectures.
- **The PDU Set Framework (WG RAN2/SA2):** Discards basic agnostic bitstream prioritization. Multimedia flows are processed at the granularity of Application Data Units (ADUs—representing full video frames/slices). RTP headers map boundaries via explicit structural metrics: Sequence Numbers (PSSN), Importance Marks (PSI), and Burst Boundaries (D) to automate informed packet discarding during air-interface saturation spikes.
- **Next-Generation Real-Time Communications (NG-RTC):** Finalized via TS 29.175/29.330, this layer embeds dynamic Service-Based Interfaces (SBI) into standard IMS infrastructure. This enables software-defined data tunnels to run text-to-speech translation engines, metaverse digital avatars, and active LLM agents directly inline within a standard calling loop.

5. TR 28.908 AI/ML Management & Operational Workflow Framework

Technical Report 28.908 governs the multi-vendor standardization guidelines required to train, validate, deploy, and manage machine learning models operating within telecommunication topologies (including NWDAF core functions and RAN intelligence loops):

A. The Standardized Operational Workflow

The model lifecycle is governed by an explicit multi-stage automated loop: **1. Training Phase:** Executes mathematical learning across a structured dataset to generate a core ML Entity. **2. Validation Phase:** Evaluates performance variance between active training weights and isolated validation subsets. **3. Testing Phase:** subjects validated candidates to entirely unseen target data profiles to compute exact compliance before field installation. **4. Deployment Phase:** Regulates the binary file distribution and environment installation within target network nodes. **5. Inference Phase:** Runs live software loops inside operational nodes to execute automated real-time local network adjustments.

B. Managed Operational Objects & Request Classes (IOCs)

To manage multi-vendor models without exposing proprietary algorithmic weights, TR 28.908 establishes standard NRM Information Object Classes (IOCs):

- **MLTrainingRequest / MLTrainingReport:** Allows an operator to configure training data properties, prescribe baseline threshold requirements, and evaluate generated performance matrices (e.g., F1 Score, Mean Absolute Error).
- **MLTestingRequest / MLTestingReport:** Creates an isolated testing run context where consumers supply independent tracking datasets to verify real-world accuracy before field activation.
- **MLEntityLoadingRequest / MLEntityLoadingPolicy:** Handles multi-vendor deployment sequences, configuring automated schedules or threshold triggers to download target binaries directly into remote nodes.
- **MLInferenceEmulationRequest:** Instantiates a software-defined digital twin sandbox environment to track behavioral model properties before unlocking active live network modification capabilities.

Advanced Orchestration Controls & Trustworthy Compliance Paradigms

6. Advanced Operational Inference & Automation Controls

To mitigate systemic operational risk within live network environments, TR 28.908 specifies granular control mechanisms governing model behavior, structural scopes, and runtime updates:

- **Granular Scope Segregation:** Models must register explicit physical or logical operational limits: *Monitoring Scope* dictates the exact boundaries of localized training data ingestion pools; *Validity Scope* establishes the specific cells or tracking areas authorized to execute inference decisions; *Prepared Scope* defines a standby mode area where models are initialized and pre-fed real-time states (e.g., LSTM state hydration) to execute instantaneous handovers without activation delay.
- **Partial & Progressive Activation Techniques:** Operators can restrict untrusted or newly deployed models through explicit *Abstract Activation Steps*. This enforces soft control caps, restricting model inference to small sub-scopes (e.g., a specific subset of test cells) or bounded temporal windows (e.g., low-traffic hours between 00:00 and 04:00) before expanding to an entire metropolitan deployment.
- **Automated Update Management & Model Drift Mitigation:** Models natively decay over time due to time-varying environmental shifts (Model Drift). The NRM framework tracks this by binding an explicit *updatedAtTime* timestamp onto the *MLEntity* data class. When internal accuracy falls below a pre-configured target, the system triggers automated re-training loops, generating a fresh version block announced via standard attribute change notifications.

7. Multi-Vendor Orchestration, Coordination & Arbitration Loops

When independent automated functions operate on shared infrastructure elements, their localized decisions can introduce severe systemic conflicts (e.g., Mobility Load Balancing trying to shift traffic out of a cell while Energy Saving attempts to deactivate it). TR 28.908 defines a centralized coordination framework to resolve these multi-vendor race conditions:

Individual functions are barred from directly altering live network parameters. Instead, they must output their requested optimizations via a standardized **recommendedAction** data type forwarded to a central **AIMLOrchestration** supervisor module. The orchestrator cross-references recommendations against a local *Capability Library*, applies concurrency locks, and selects the optimal path. Following execution, the orchestrator triggers adjacent peer nodes to enter a dedicated monitoring window. These peer nodes calculate localized system impacts and return an abstract, standardized **Action Quality Indicator (AQI)** metric. By analyzing the aggregate AQI, the orchestrator detects multi-vendor interference patterns, bars conflicting parameters, and automatically rescales the operational boundaries of errant models.

8. The Three Pillars of Trustworthy AI Compliance

To comply with emerging global regulatory guidelines (such as the European Union AI Act and ISO/IEC TR 24028), the 3GPP management system formalizes data modeling parameters around three core Trustworthy Machine Learning pillars:

Trustworthy Pillar	Standard Metric Definition	Algorithmic Enforcement Action
1. Explainability	Local & Global Explanations; Monotonicity Evaluation Indices.	Computes feature importance rankings to trace exactly how an inference was reached without exposing raw underlying source code.
2. Fairness	Disparate Impact Ratio; Average Odds Difference Matrix.	Evaluates selection ratios between privileged and unprivileged groups to strip out bias (e.g., blocking uneven cell deactivation patterns).
3. Robustness	Missingness Ratio; Adversarial Distortion Bounds.	Forces mean-value feature imputation and injects mathematical noise profiles into training inputs to guard against poisoning injection attacks.

Footnote: This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.